

Topic 1.7

1.7: Passing Arrays to Functions

An important consequence of all this is that arrays appear to be passed to functions by reference. This really isn't true, but that's the common perception.

```
int max( int x[], int n )
{
    int maximum=x[0];
    int i;

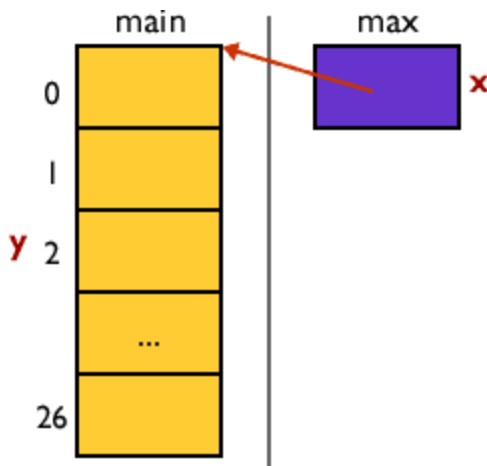
    for ( i=1; i<n; ++i)
        if (x[i] > maximum)
            maximum = x[i];
    return maximum;
}

int main(void)
{
    int y[27], largest;

    largest = max(y, 27); // Passes address of y[0].

    largest = max( &y[1], 26 ); // Explicitly passes
                                // address of y[1].

    return 0;
}
```



In the above declaration of the `max` function, the first formal parameter declaration `int x[]` tells the compiler that the first actual parameter should be a pointer to an `int`, which is (hopefully) the first element in some array. This may be changed to

```
int max( int *x, int n )
```

with no other changes required, and has exactly the same meaning as far as the compiler is concerned. Use whichever declaration is more informative to the human reader.

Pointers and arrays are effectively the same, except that *declaring an array variable forces memory to be allocated for all the elements in the array variable*, whereas declaring a pointer variable causes memory to be allocated for the pointer only. It is sometimes more convenient or intuitive to use the pointer notation, and sometimes more appropriate to use the array notation.

Pointers to Functions

The name of a function actually refers to the address at which the function's compiled code begins. It is also useful to have *pointers* to functions. Consider:

```
int mine( float x )
{
    ...
}

int yours( float x )
{ ...}

int main()
{
    int (*funcp) (float);
    int result;

    funcp = mine;
    result = funcp(3.4);
    result = (*funcp)(3.4);    /* Alternate invocation form. */
    return 0;
}
```

The declaration of *funcp* declares it to be a pointer to a "function that takes a float argument and produces an *int* result."

to topic 1.8 (<https://onlinelearning.berkeley.edu/courses/2291660/pages/topic-1-dot-8>)